

Wheel Balancing

truing of grinding wheels. The grinding wheels are balanced by mounting them on test mandrels.

The wheel along with the mandrel is rolled on knife edges to test the balance and corrected.

4.15 Cutting speed, Feed and depth of cut

Cutting speed

Cutting speed of a grinding process is the relative speed of the grinding wheel and the workpiece. It is expressed in metre/minute.

Cutting speed (c.s) = πDN /1000 metre/min

π = 22/7 (or) 3.14

D = The diameter of the grinding wheel in mm

N = The speed of the grinding wheel in RPM.

Feed in a grinding process is the longitudinal movement of the work mounted on the table per revolution of the grinding wheel. It is express in mm per revolution.

The longitudinal feed during rough grinding is approximately 0.6 to 0.9 of the width of the wheel and 0.4 to 0.6 of the width of the wheel during finish grinding.

Depth of cut

The thickness of the metal layer removed from the work in one pass of the wheel is known as depth of cut. It is expressed in mm. Depth of cut is kept ranging from 0.005 to 0.04 mm

4.16 Precision Operations

In a workshop, metal parts are manufactured by performing different operations in lathe, shaping machine, milling machine, drilling machine or grinding machine. In order to enhance the quality of surfaces of these parts, several surface finishing processes are performed on them. However, if a better finish is desired for looks, for accuracy, for wearing qualities or for any other reasons, one of the following processes is employed.

1. Lapping
2. Honing
3. Super finishing
4. Polishing
5. Buffing
6. Scraping
7. Electroplating

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